

HEINO, Netherlands

WMO: 062780

Lat: 52.434N

Lon: 6.259E

Elev: 5

StdP: 101.26

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.2	-5.2	-10.2	1.6	-5.8	-7.8	1.9	-4.5	11.3	9.7	10.2	9.4	2.4	70	0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	28.7	19.9	26.4	19.2	24.5	18.2	21.1	26.7	20.1	25.1	19.1	23.2	3.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.2	14.0	23.3	18.2	13.1	22.1	17.4	12.5	21.1	61.1	27.0	57.5	25.1	54.3	23.4	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.8	7.6	6.6	-9.9	33.2	3.4	2.1	-12.4	34.7	-14.4	35.9	-16.3	37.0	-18.8	38.5		
(5)			-10.3	23.8	3.4	1.2	-12.7	24.6	-14.7	25.3	-16.6	25.9	-19.0	26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	2.9	3.6	5.9	9.4	13.0	15.8	17.7	17.3	14.5	10.8	6.8	3.7
	DBStd	6.33	4.21	3.92	3.56	3.54	3.54	3.02	2.96	2.61	2.78	3.42	3.37	4.36
	HDD10.0	933	222	181	133	54	11	0	0	0	1	32	104	196
	HDD18.3	3051	479	413	386	269	169	87	48	52	118	233	345	453
	CDD10.0	992	1	1	6	35	104	174	239	226	138	57	9	2
	CDD18.3	69	0	0	0	0	4	12	29	19	4	1	0	0
	CDH23.3	686	0	0	0	9	61	134	275	167	38	1	0	0
	CDH26.7	175	0	0	0	1	9	33	82	44	6	0	0	0
(14) Wind	WSAvg	3.1	3.8	3.7	3.5	3.0	3.0	2.7	2.7	2.5	2.5	3.0	3.1	3.7
(15) Precipitation	PrecAvg	846	73	62	55	45	62	72	91	87	73	73	75	77
	PrecMax	1135	131	123	116	95	124	156	209	215	166	212	148	170
	PrecMin	631	9	19	11	0	27	13	26	14	8	25	10	23
	PrecStd	124	34	30	32	22	25	33	38	49	39	36	31	35
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.9	14.3	18.2	24.8	27.8	30.5	32.4	31.0	27.2	21.8	16.1	13.5
		MCWB	11.1	9.9	12.2	15.6	18.3	20.8	21.0	21.0	20.0	17.1	13.8	11.7
	2%	DB	11.2	11.7	15.3	21.1	24.9	26.9	28.9	27.5	24.1	18.8	14.3	12.2
		MCWB	9.8	9.4	10.8	14.1	16.4	19.1	20.2	19.9	18.8	15.7	12.7	10.9
	5%	DB	9.8	10.4	13.0	18.5	22.7	24.4	26.3	24.9	21.7	17.0	13.0	10.9
		MCWB	8.6	8.6	9.9	12.8	15.7	18.2	19.5	18.9	17.4	14.5	11.4	9.7
	10%	DB	8.5	9.0	11.4	16.2	20.2	22.1	24.1	22.7	19.6	15.6	11.7	9.5
		MCWB	7.4	7.5	8.9	11.8	14.7	17.0	18.5	17.9	16.1	13.7	10.4	8.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.4	11.1	13.0	16.5	19.6	22.0	22.7	22.3	20.6	17.6	14.3	12.2
		MCDB	12.4	12.6	16.1	22.8	25.8	28.0	28.6	29.0	25.7	20.9	15.4	12.9
	2%	WB	10.1	9.9	11.6	14.8	17.9	20.2	21.2	20.7	19.2	16.0	13.0	11.0
		MCDB	11.0	11.2	14.3	20.2	22.9	25.2	27.0	25.7	23.3	18.4	14.3	12.1
	5%	WB	8.7	8.7	10.4	13.4	16.4	18.9	20.2	19.5	17.9	15.0	11.7	9.8
		MCDB	9.6	10.2	12.8	17.6	21.2	23.1	25.4	23.9	21.1	16.7	12.9	10.9
	10%	WB	7.5	7.6	9.2	12.1	15.1	17.6	19.1	18.5	16.7	14.0	10.5	8.5
		MCDB	8.5	9.0	11.1	15.6	19.3	21.3	23.0	21.9	19.1	15.4	11.6	9.4
(35) Mean Daily Temperature Range	5% DB	MDBR	4.7	5.8	7.7	10.1	10.1	9.8	9.4	9.5	9.0	7.2	5.3	4.5
		MCDBR	4.9	7.3	11.1	14.4	14.2	13.9	13.9	13.6	12.2	8.4	5.8	4.7
		MCWBR	4.5	5.5	7.4	8.1	7.2	7.2	6.6	7.1	7.2	5.6	4.7	4.2
	5% WB	MCDBR	5.1	6.5	9.9	12.8	12.4	12.0	12.3	11.9	10.8	8.0	5.6	4.8
		MCWBR	4.7	5.3	6.9	7.7	6.9	7.0	6.3	6.6	6.7	5.7	4.8	4.5
(40) Clear-Sky Solar Irradiance	taub		0.329	0.351	0.380	0.408	0.411	0.408	0.420	0.412	0.389	0.364	0.343	0.324
	taud		2.331	2.325	2.255	2.223	2.253	2.275	2.252	2.294	2.350	2.389	2.354	2.320
	Ebn at Noon		652	751	800	824	841	846	826	811	785	726	628	585
	Edh at Noon		63	83	108	126	128	127	128	116	98	78	61	54
(44) All-Sky Solar Radiation	RadAvg		0.68	1.35	2.51	4.06	4.96	5.26	5.03	4.21	3.05	1.76	0.80	0.52
	RadStd		0.07	0.24	0.32	0.42	0.41	0.38	0.61	0.35	0.33	0.17	0.08	0.06

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page